

1. PATH GAME BASED

Game Concept & Rules

Objective: Rotate tiles to create a continuous path from start to end

Core Mechanics:

- Each tile has path segments in specific directions
- Rotate tiles 90° at a time
- Path must be continuous and match directions
- All tiles don't necessarily need to be used
- | is bi directional up and down

S->		0	->
0	->		
0	0		<-
0	0	->	
<-	0		E

Step 1 ->

Go to (0,1)

S->		0	->
0	->		
0	0		<-
0	0	->	
<-	0		E

Step 2 :

Go to (1,1)

S->		0	->
0	->		
0	0		<-
0	0	->	
<-	0		E

Step 3 :

Rotate 4 and Go to (1,2)

S->		0	->
0	->	->	
0	0		<-
0	0	->	

<-	0		E
----	---	--	---

Step 4 :

Go to (1,3)

S->		0	->
0	->	->	
0	0		<-
0	0	->	
<-	0		E

Step 5 :

Rotate 3 times and Go to (2,3)

S->		0	->
0	->	->	
0	0		
0	0	->	
<-	0		E

Step 5 :

Go to (3,3)

S->		0	->
0	->	->	
0	0		
0	0	->	
<-	0		E

Step 5 :

Go to (3,4)

S->		0	->
0	->	->	
0	0		
0	0	->	
<-	0		E

Example Level 2: With Rotation Tiles

Initial Board:



Step-by-Step Solution:

Step 1: Analyze available tiles

- Movement: →, →, ↓, →, →
- Rotation: ↻ (turn right)
- Must use ↻ to navigate around wall

Step 2: Plan optimal path

text

Need to go: Right → Right → Turn down → Down → Right → Right → Reach X

Step 3: Rotate tiles strategically

Keep tiles as is! The path works:

Execution:

[0,0] Start facing right

[0,1] Move right (on → tile)

[0,2] Move right (on ↻ tile - turn right, now facing down)

[1,2] Move down (on ↓ tile)

[2,2] Move right (on → tile)

[2,3] Move right (on → tile) - Reach X ■

Step 4: Visualize execution:

Step 1:

o	R→	→	o
o	█	↑	o
o	→	→	X

Step 2:

o	o	R→	o
o	█	↑	o
o	→	→	X

Step 3: (after ↑ rotation, now facing down)

o	o	o	o
o	█	R↓	o
o	→	→	X

Step 4-6: Continue to X...

2.MATHS BASED GAMES

You will get total 25 games like this in exam , in which you need to choose correct order

Instructions

For each game below, compare the three given expressions. Arrange them in **ascending order** and explain your reasoning. Watch out for equal values and subtle decimal differences!

Game 1

Options:

- $12 / 5$
- 1.2
- 2.3

Correct Order:

$$1.2 < 2.3 < 2.4$$

Explanation:

$12 \div 5 = 2.4$, which is greater than 2.3.

Game 2

Options:

- 3.5
- $7 / 2$
- 3.25

Correct Order:

$$3.25 < 3.5 = 3.5$$

Explanation:

$7 \div 2 = 3.5$, equal to the decimal 3.5.

Game 3

Options:

- $5 / 4$
- $1.1 + 0.2$
- $1.6 - 0.1$

Correct Order:

$$1.25 < 1.3 < 1.5$$

Explanation:

$5 \div 4 = 1.25$, $1.1 + 0.2 = 1.3$, and $1.6 - 0.1 = 1.5$.

Game 4

Options:

- 2.9
- $3 - 1/5$
- $14 / 5$

Correct Order:

$$2.8 = 2.8 < 2.9$$

Explanation:

$3 - 1/5 = 2.8$, $14 \div 5 = 2.8$. Both are equal and less than 2.9.

 Game 5

Options:

- $\sqrt{9}$
- 2.8
- $3.1 - 0.1$

Correct Order:

$$2.8 < 3.0 = 3.0$$

Explanation:

$\sqrt{9} = 3$, and $3.1 - 0.1 = 3$.

 Game 6

Options:

- $8 / 3$
- $2.5 + 0.1$
- 2.75

Correct Order:

$$2.6 < 2.666 < 2.75$$

Explanation:

$8 \div 3 \approx 2.666$, and $2.5 + 0.1 = 2.6$.

 Game 7

Options:

- 1.9×1.1
- 2.05
- $19 / 9$

Correct Order:

$$2.05 < 2.09 < 2.11$$

Explanation:

$1.9 \times 1.1 = 2.09$, $19 \div 9 \approx 2.111$.

 Game 8

Options:

- $15 / 7$
- 2.1
- $2.05 + 0.05$

Correct Order:

$2.1 = 2.1 < 2.14$

Explanation:

$15 \div 7 \approx 2.142857$, which is greater than 2.1 .

 Game 9

Options:

- $2^3 / 5$
- $3.3 - 1$
- $7 / 2$

Correct Order:

$1.6 < 2.3 < 3.5$

Explanation:

$2^3 = 8 \rightarrow 8 \div 5 = 1.6$, $3.3 - 1 = 2.3$, $7 \div 2 = 3.5$.

 Game 10

Options:

- $\sqrt{16} / 2$
- 2.1
- $4 / 1.5$

Correct Order:

$2.0 < 2.1 < 2.66$

Explanation:

$\sqrt{16} = 4 \rightarrow 4 \div 2 = 2$, and $4 \div 1.5 \approx 2.666$.

3.THE HIDDEN PATH GAME

Game Rules

1. **Goal:**
Start from S (Start) → collect the Key (K) → reach the Exit (E) safely.
2. **Hidden Walls:**
Some cells have invisible walls (X) — you can't see them until you try to move into them.
3. **Movement:**
 - You can move Up, Down, Left, or Right (no diagonal moves).
 - Each move reveals if the new cell is safe or blocked.
4. **Blocked Cell:**
 - If you hit a wall (X), you stay in your current cell and mark that position as blocked.
 - Try another direction.
5. **Key Rule:**
 - You must collect the Key (K) before the Exit (E) opens.
 - If you reach E without the key, the door won't open.
6. **Winning Condition:**
 - You win when you reach E (Exit) *after collecting the key*.
7. **Losing Condition (optional):**
 - You lose if you get trapped (no open moves left).
 - Restart the level and try a new path.

Maze layout (visual)

Rows numbered 1→5 (top→bottom), columns 1→5 (left→right).

C1 C2 C3 C4 C5

R1 [S] [.] [.] [.] [.]

R2 [.] [X] [.] [X] [.]

R3 [.] [.] [K] [.] [.]

R4 [X] [.] [X] [.] [.]

R5 [.] [.] [.] [X] [E]

Legend

- S = Start at (1,1)

- K = Key at (3,3)
- E = Exit/Door at (5,5)
- X = invisible wall (unknown until you hit it)
- . = unknown/open cell

Walls (hidden to player until hit): (2,2), (2,4), (4,1), (4,3), (5,4).

Goal

1. Reach **K (3,3)** and collect the key.
2. Then reach **E (5,5)**.
You must **avoid invisible walls** — mark them when discovered.

Step-by-step playthrough (with discovery + final successful route)

Attempt 1 — exploratory (you will *discover* a wall)

1. Start at **(1,1)**. Move **Right** → go to **(1,2)**. (Safe)
 - Map note: mark (1,2) as safe.
2. From **(1,2)** try **Down** → attempt to go to **(2,2)**.
 - **Thud! Invisible wall at (2,2)**. Mark (2,2) as blocked.
 - Back up to **(1,2)** (you cannot stay in a blocked cell).
3. Having found a wall at (2,2), you decide to change strategy (don't try to go down from (1,2) again).

Plan update (memorize)

- Known blocked cells: **(2,2)**
- Known safe: **(1,1), (1,2)**
- New **plan**: try going **down from start** instead of forcing through (2,2).

Successful route to the key (recorded and memorized)

From **Start (1,1)** follow these moves:

4. **Down** → (2,1) — safe.
5. **Down** → (3,1) — safe.
6. **Right** → (3,2) — safe.
7. **Right** → (3,3) — **Key found!**

So the directions from start to key: **D, D, R, R**
(That's 4 moves, learned after discovering the blocked (2,2) cell.)

From Key to Door (safe route)

With the key at (3,3), continue:

8. **Right** → (3,4) — safe.
9. **Down** → (4,4) — safe.
10. **Right** → (4,5) — safe.
11. **Down** → (5,5) — **Door reached** (key collected → door opens).

So directions from key to door: **R, D, R, D** (4 moves)

Final combined minimal route (after mapping)

Start → Key → Door with the discovered safe path:

- From (1,1) to (3,3): **D, D, R, R**
- From (3,3) to (5,5): **R, D, R, D**

Full move sequence: **D, D, R, R, R, D, R, D**

Total *successful* moves after mapping: **8 moves** (plus the earlier failed attempt that discovered (2,2)).

Visual: final path with step numbers

(Shows the successful route and marks discovered walls)

```
C1 C2 C3 C4 C5
R1 [ S ] [ . ] [ . ] [ . ] [ . ]
R2 [ 1 ] [ X ] [ . ] [ X ] [ . ]
R3 [ 2 ] [ 3 ] [ K ] [ 5 ] [ . ]
R4 [ X ] [ . ] [ X ] [ 6 ] [ 7 ]
R5 [ . ] [ . ] [ . ] [ X ] [ E ]
```

Numbers indicate the step order to reach the Key (1–4) and then door (5–8):

- Step1: (2,1)
- Step2: (3,1)
- Step3: (3,2)
- Step4: (3,3) KEY

- Step5: (3,4)
- Step6: (4,4)
- Step7: (4,5)
- Step8: (5,5) EXIT

Practice game

Game Setup (Visualization):

Imagine a **5x5 maze grid** like this:

```
[S][R][ ][G][ ]
[ ][ ][B][ ][ ]
[ ][Y][ ][ ][ ]
[ ][ ][ ][ ][D]
[ ][ ][ ][ ][ ]
```

Legend:

- S = Start
- R = Red switch
- G = Green switch
- B = Blue switch
- Y = Yellow switch
- D = Door (Exit)

Goal:

- You must **activate all switches in the correct color order** ($R \rightarrow G \rightarrow B \rightarrow Y$).
 - After all switches are activated in order, the **Door (D)** opens.
 - Then move to the Door to win.
-

Rules:

1. **Start at S (Start).**
2. You can move **up, down, left, right** — one block per turn.

3. There are **invisible traps** — stepping on them sends you **back to Start**.
 4. You must **activate the switches in the correct sequence**:
Red → Green → Blue → Yellow.
 5. If you step on a wrong color before completing the sequence, all switches reset.
 6. Once all four switches are activated, the **door (D)** becomes active.
 7. Reach D to **win the game**.
-

Example Gameplay (Step-by-Step Visualization):

- Start at S.
- Move **right** → reach R (Red Switch) → Activated!
- Move **right, right** → reach G (Green Switch) → Activated!
- Move **down, left** → reach B (Blue Switch) → Activated!
- Move **down** → reach Y (Yellow Switch) → Activated!
- Move **right, right** → reach D (Door) → Opens → **You win!**

4.MEMORY GAME

Example Game Grid (5×5)

(You don't know where X walls are until you hit them.)

Actual hidden layout:

S	.	.	X	.
.	X	K	.	.
.	.	X	.	.
.	X	.	E	.
.	.	.	.	.

What you see at the start:

S ? ? ? ?

? ? ? ? ?

? ? ? ? ?

? ? ? ? ?

? ? ? ? ?

Gameplay Example

Step 1: Start at S (0,0).

Try **Right** → (0,1) is safe .

Grid now:

text

S . ? ? ?

? ? ? ? ?

? ? ? ? ?

? ? ? ? ?

? ? ? ? ?

Step 2: From (0,1), try **Right** → (0,2) is safe .

Grid now:

S . . ? ?

? ? ? ? ?

? ? ? ? ?

? ? ? ? ?

? ? ? ? ?

Step 3: From (0,2), try **Right** → (0,3) is **X** (blocked)!

You bounce back to (0,2) and mark (0,3) as X:

text

S . . X ?

? ? ? ? ?

? ? ? ? ?

? ? ? ? ?

? ? ? ? ?

Step 4: From (0,2), try **Down** → (1,2) is safe and has **K** (Key)!

You collect the key. Grid now:

S . . X ?

? ? K ? ?

? ? ? ? ?

? ? ? ? ?

? ? ? ? ?

Step 5: Now find the Exit E.

From (1,2), try **Down** → (2,2) is **X** (blocked).

Mark it:

S . . X ?

? ? K ? ?

? ? X ? ?

? ? ? ? ?

? ? ? ? ?

Step 6: From (1,2), try **Right** → (1,3) is safe .

Then **Down** from (1,3) → (2,3) is safe .

Then **Down** from (2,3) → (3,3) is safe and has **E** (Exit).

Since you have the key, you **WIN**.

Final Discovered Map

S . . X .

. X K . .

. . X . .

. X . E .

.

Code bashers

5.Hidden Layout (6×6)

Actual Hidden layout:



Player's starting view:

```
S?????  
??????  
??????  
??????  
??????  
??????
```

Gameplay Steps

Step 1: Start at S (0,0).

Try Right → (0,1) is safe .

Grid now:

```
S.????  
??????  
??????  
??????  
??????  
??????
```

Step 2: From (0,1), try Right → (0,2) is X (blocked).

Mark it:

S . X ? ? ?

? ? ? ? ?

? ? ? ? ?

? ? ? ? ?

? ? ? ? ?

? ? ? ? ?

Step 3: From (0,1), try Down → (1,1) is X (blocked).

Mark it:

In the Mind

In memory

S . X ? ? ?

. X ? ? ? ?

? ? ? ? ?

? ? ? ? ?

? ? ? ? ?

? ? ? ? ?

S ? ? ? ? ?

? ? ? ? ?

? ? ? ? ?

? ? ? ? ?

? ? ? ? ?

? ? ? ? ?

Step 4: From (0,1), only Left is start, so go Down not possible, Right not possible — wait, we must go Down from (0,1)? Already blocked. So backtrack: actually from (0,1), we can only go Down (tried, blocked) or Right (blocked), so we must go Left back to start (0,0). Then from (0,0), try Down → (1,0) is safe .

Grid now:

S . X ? ? ?

. X ? ? ? ?

? ? ? ? ?

? ? ? ? ?

? ? ? ? ?

? ? ? ? ?

Step 5: From (1,0), try Right blocked (1,1 is X), try Down → (2,0) is safe .

Grid:

S . X ? ? ?

. X ? ? ? ?

. ? ? ? ? ?

? ? ? ? ? ?

? ? ? ? ? ?

? ? ? ? ? ?

Step 6: From (2,0), try Right → (2,1) is safe .

Try Right again → (2,2) is safe and has K (Key)! Collect it.

Grid:

S . X ? ? ?

. X ? ? ? ?

. . K ? ? ?

? ? ? ? ? ?

? ? ? ? ? ?

? ? ? ? ? ?

Step 7: Now find Exit.

From (2,2), try Right → (2,3) is X.

text

S . X ? ? ?

. X ? ? ? ?

. . K X ? ?

? ? ? ? ? ?

? ? ? ? ? ?

? ? ? ? ? ?

Step 8: From (2,2), try Down → (3,2) is X.

Mark it:

S . X ? ? ?

. X ? ? ? ?

. . K X ? ?

??X???

??????

??????

Step 9: From (2,2), try Up → (1,2) is safe .

From (1,2), try Right → (1,3) is safe .

From (1,3), try Right → (1,4) is X.

Mark it:

S . X ???

. X . . X ?

. . K X ? ?

??X???

??????

??????

Step 10: From (1,3), try Down → (2,3) is X, try Up from (1,3) → (0,3) is safe .

From (0,3), try Right → (0,4) is safe .

From (0,4), try Down → (1,4) is X, try Right → (0,5) is safe .

From (0,5), try Down → (1,5) is safe .

From (1,5), try Down → (2,5) is safe .

From (2,5), try Down → (3,5) is safe .

From (3,5), try Down → (4,5) is safe and has E (Exit)!

Since we have key, we WIN.

Final Discovered Map

S . X . . .

. X . . X .

. . K X . .

??X???

?????E

?????.

(Some cells in rows 3–5 remain unexplored but we found the exit.)